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A Review of Agentic Artificial Intelligence: Power of Self-Driven AI in the Future of Financial Autonomy and Enhanced Customer Engagement

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***Abstract*—The evolution of Large Language Models, Robo-advisors or chatbots, and Machine Learning Models have opened the possibilities for automation across sectors to innovate and introduce new capabilities as part of their business process. This review paper targets all the critical aspects of artificial intelligence and machine learning models and their impact on the successful deployment of the Agentic AI capability. Agents can not only participate but also assist, self-organize, and negotiate using parallel workstreams and cognitive architectures. They solve problems iteratively based on their cognitive level and domain, creating value in the financial services market, where opportunities are vast. This study further includes principles for Agentic AI, its lifecycle, various key components, impact on the financial industry, architectural considerations, data considerations, benchmarking, opportunities, future capabilities, challenges, and more.**

***Index Terms*—Generative Artificial Intelligence (Gen AI), Agentic Artificial Intelligence (Agentic AI), Finance, Artificial Intelligence Architecture, Large Language Model (LLM), Machine Learning, Automation, Customer Experience, ChatGPT, Natural Language Processing (NLP), Reinforcement Learning**

I. Introduction

Artificial Intelligence initiatives and use cases are continuing to increase in the Financial Services Industry and has potentially contributed to new experiments that will enable the user experience. With increased number of experiments, the adoption across the organization or industry is a big challenge. Data security, LLM reliability, and regulatory oversight concerns for AI-powered financial decisions represent major stumbling blocks. Although most AI applications in finance function internally today, customer-facing chatbots handle noncritical service requests. Currently, there is a growing interest in gen AI agents, referred to as "Agentic AI," where LLMs are used to autonomously consume, produce, and interact through natural language.

With the evolution of AI technology, Agentic AI is set to revolutionize the financial sector like never before. In contrast to conventional generative AI, which only provides outputs based on human inputs, Agentic AI is capable of observation, reasoning, direct action, and even independent learning without needing human input all the time. A study conducted in 2020 by the Cambridge Centre for Alternative Finance [19] and the World Economic Forum [12] emphasized the growing significance of AI in the finance sector. With the power of generative AI now extended, Agentic AI will bring the financial service industry closer to operational independence with faster data processing, better decision accuracy, personalized customer interactions, and the ability to adapt in real time to market changes. The financial industry is the second-largest user of regenerative AI, behind telecom and media. Generative and Agentic AI should only fuel this trend, given that historically, banks have been one of the biggest technology investors (outside of tech companies) [1].

II. The Economic Impact of Gen AI And The Development of MultiModal Agentic Framework

Generative AI (GenAI) is primed to initiate major economic change, with conservative estimates suggesting a Global GDP contribution between $2.6 trillion and $4.4 trillion per year across the industries by 2030. By the end of the decade, GenAI investments in energy alone will triple, climbing from $40 billion in 2023 to $140 billion+. This rise brings out GenAI's transformational power in improving productivity, optimizing business processes, and reimagining value chains [12].

Amidst this fast-paced evolution, multimodal GenAI agentic frameworks have emerged as one of the key catalysts for supercharging process automation across enterprises to an unprecedented scale. Combining many individual specialized agents within large uncased multi-task systems creates powerful, well-defined systems that are integrated with unknown layers for innovation to solve defined problems. Multimodal agents can leverage various AI methods to handle data from different modalities (including text, images, and audio), facilitating improved decision-making, process improvement, and autonomous activity performance.

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A. The Rise of Multimodal GenAI Agents

Multimodal Generation Artificial Intelligence agents is an extension of early agentic AI concepts. The multimodal being an extension is more advanced with respect to the logic and rules, and is advanced to manage better human-like interactions. The three major phases are [7]:

*1) **Integration of Machine Learning (2000–2010)**:* (i) Data-driven learning: Machine Learning concepts can be integrated to AI agents and this integration will facilitate learning from large datasets allowing the application to make better decision which is technically difficult to implement using the traditional rules engine methodology, and (ii) Advancements in Natural Language Processing (NLP): Enhanced NLP capacities that enabled AI to understand and generate human language more effectively, making interactions more natural and efficient.

*2) **Emergence of Multimodal AI (2010–2020)**:* (i) Cross-modal integration: AI agents matured to process and integrate multiple modalities of information—text, images, and audio—into actionable insights. The multimodal concepts keeps evolving and the most important of use cases includes the identification of objects on images and pictures, and building comprehension as part of the voice based files or recordings, and (ii) Enhanced user interactions: It is important aspect of Agentic AI to improve the user experience, user engagement and making more contextual and improved understanding of the data and its value while serving the user.

*3) **Advanced Autonomy and Real-Time Interactions (2020–Current)**:* (i) Autonomous decision-making: Agentic AI can independently set goals, plan to achieve them, and execute commands in real time. Orchestration in multi-agent systems (MAS) improves slow but thorough reasoning (ChatGPT-1o). When combined with fast (ChatGPT-4o) thinking for solution generation, it leads to high validation of creative ideas. Agentic AI is a concept that creates opportunities to independently perform actions and achieve tasks which is not possible in traditional systems. Few of the examples or use cases that uses Agentic AI concepts include the self driving feature in cars, automated business processes and chotbot implementation which has a real-time customer support features, and (ii) Ethical and responsible AI governance: the global environment and development has continuously changed with next-generation technology and it is important to make ethical and moral considerations while using the data and output from any Artificial Intelligence systems and solutions.

III. What is Agentic AI?

Agentic Artificial Intelligence are agents that have autonomy in their behavior rather than just responding retroactively to the user's command, as is typical in people following lines of AI. Traditional AI fetches or generates responses, Agentic AI decides on actions, orchestrates multi-step workflows, and makes real-time adaptations. These systems combine LLMs, RL, RAG, and multi-agent systems to delegate large-scale, high-level discussions with limited human supervision. These organizations are not only automating specific rules-based processes anymore but also augmenting human decision processes to provide them with insight when they make financial decisions with Agentic AI [2].

Agentic AI is characterized by three core capabilities that are: (i) **Autonomy:** The ability to make decisions, perform actions, and optimize via strategies without needing continuous human input, (ii) **Adaptability:** The ability to learn interactions, market movement patterns, and new data over time to improve predictions and decision-making, amd (iii) **Coordination:** The ability to communicate with other AI agents, APIs, and financial databases to execute more extensive operations like portfolio rebalancing and fraud detection. By integrating these capabilities, Agentic AI is revolutionizing financial services, shifting institutions from rule-based automation to intelligent decision augmentation, enhancing efficiency, risk management, and personalized customer experiences [6].

A. What Can AI Agents Do?

Agentic AI employs its "chaining" capabilities, which allow it to perform a chain of related tasks in response to a single query. This enables AI Agents to [9]: (i) Self-assess scenarios and decide on the best pathway to progress, (ii) Gain wisdom from prior successes and failures to improve future decision-making, (iii) Execute elaborate, multi-layered processes independently and without human mediation, and (iv) Adjust strategies to live according to the evolving conditions.

Agentic AI empowers a very high level of autonomy, which is critical in financial operations where speed, accuracy, personalized outcomes, and compliance with the applicable regulations are paramount. While traditional AI acts mostly as an assistant to human decision-makers, Agentic AI takes matters into its own hands, running entire workflows end-to-end, from analysis to execution to monitoring. Early applications of this technology are Microsoft Copilot, an AI-powered assistant, and Google AI Teammate, which independently manages projects by observing company processes [8].

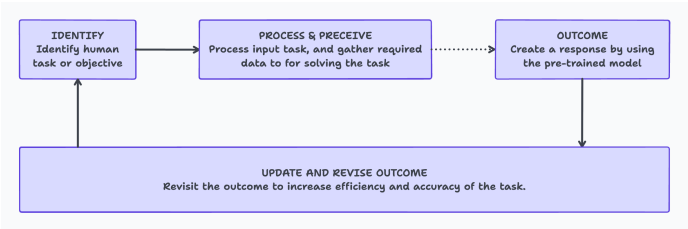

Fig. 1. Lifecycle Illustration for Generative Artificial Intelligence [19].

B. Why Finance Needs AI That Thinks and Acts Independently

Financial Industry is going through a major transition with all the technological advancement, and one such extension of technology is Agentic AI. It is important to note that the data is expanding at a rapid rate, and this makes it even more important for financial industry to make real-time and independent decision based on available data with less dependency on traditional rules engine, while keeping the response accurate and the response time with reduced latency.

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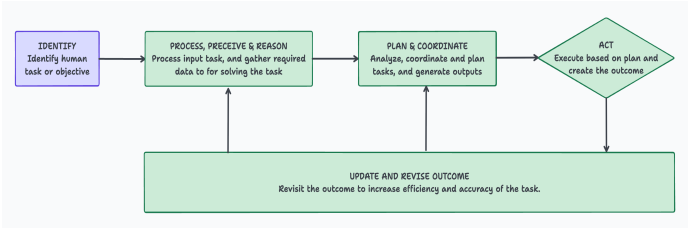

Fig. 2. Human-Like Agentic Artificial Intelligence Lifecycle Illustration [19].

Traditional decision-making models, even those enhanced by conventional AI, struggle to keep pace with these challenges. Agentic AI overcomes these limitations by enabling continuous, autonomous decision-making that scales with organizational needs while maintaining consistency, accuracy, and regulatory compliance [15]. Details of the Use cases are covered as part of Table I.

IV. Core Components of Agentic AI in Finance

Agentic AI in finance operates through four key layers, each enabling autonomous decision-making, execution, and continuous improvement [11]. Details of the components are described in Table II.

V. Methodologies in Agentic AI Development

A. Architectural Approaches

Mimicking cognitive principles, the architectural designs of Agentic AI usually follow modular and hierarchical frameworks that can allow for systems to manage complex objectives while enabling dynamic adaptation to changing environments. The most prevailing architectures are comprised of Multi-Agent Systems (MAS), Hierarchical Reinforcement Learning (HRL), and Goal-Oriented Modular Architectures [17].

Multi-Agent Systems (MAS): In MAS, multiple autonomous agents work together or compete to complete a shared goal. The emergence of the agent-based architecture is quite beneficial if the complicated goals can be divided into simple goals, enabling the agents to specialize in different roles.

Hierarchical Reinforcement Learning (HRL): HRL hierarchically organizes decision-making, with high-level agents specifying sub-goals and low-level agents achieving them. Well-suited for multi-layered task management, maximized efficiency, and decision complexity. This design leverages the experience you gained in handling a multitude of different tasks, along with decision-making steps and the time it takes, etc. [3].

Goal-Oriented Modular Architectures: These architectures use goal-oriented structured approaches to use specialized modular agent components to each of which is focused on an isolated part of the task. This modularity allows for increased flexibility and scalability, enabling the agents to dynamically reconfigure by the demands of the task [6].

B. Advancements in Methodologies

Agentic AI methodology provides features that are vital for the financial industry as the features provide technical advantage to stay competitive and manage user experience at a level that is less possible or sometimes challenging with traditional ways. These details are as listed and explained below [2]: (i) Reasoning and Planning: Advanced framework empowers AI agents to predict outcomes, prioritize action items, and adjust strategies dynamically. Such capabilities are critical for complex, multi-objective tasks in dynamic settings like disaster response and autonomous navigation, (ii) Tool Usage and Integration: Agents integrated with external tools and APIs avail themselves to real-time computations, live data calls, and scenario simulations, enhancing decision-making accuracy and flexibility tremendously [6], (iii) Memory Mechanisms: Capable of being agentic with the use of episodic and semantic memory models to invoke context, leading to the thought of recollection of past interactions, and aims to optimize the ongoing task [6], (iv) Retrieval-Augmented Generation (RAG): RAG enables AI agents to access external knowledge on the fly, which leads to greater relevance, accuracy, and contextual depth in their outputs. One extremely powerful application is conversational AI and real-time decision-making, for instance, and (v) Instruction Fine-Tuning: Fine-tuning teaches AI agents to interpret and execute nuanced directives, empowering them to undertake multi-step tasks with precision, adaptability, and efficiency [3].

C. Benchmarking

Agentic AI is a new technology and it is critical to evaluate the possible benchmarking options to align the output is consistent and the results are matching the expectations. The below datasets and environments are useful to benchmark Agentic AI features [13]: (i) **Financial Data:** Publicly accessible data available from NASDAQ, Bloomberg and Yahoo finance systems can be utilized to train the Agentic AI model to make the decision more robust, accurate and predictable. This approach will also make the prediction engine reliable and creates confidence for the user experience, and (ii) **Healthcare Datasets:** Examples include MIMIC-III and PhysioNet. These datasets are commonly used to train and validate AI-driven patient monitoring, clinical diagnostic prediction, and decision support systems. Achieving high performance on healthcare benchmarks shows an AI system's capacity to manage sensitive, life-critical information both reliably and accurately. Benchmarking confirms that Agentic AI satisfies industry-specific requirements and enables enhanced trust, scalability, and adoption across high-stakes domains.

D. Critical Evaluation of Existing Implementation

As a continuation to what we reviewed on benchmarking of Agentic AI it is important to weigh in the success scenarios and limitations of Agentic AI's implementation, and in this section we review the details [4].

The **Successes** of Agentic AI in different industries are: (i) **Finance:** Trading is one of the use cases that will benefit from

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TABLE I
AGENTIC AI USE CASES AND APPLICATIONS IN THE FINANCIAL SERVICES DOMAIN [15]

Use Case	*Wealth Management and Retail Banking*	*Insurance*	*Institutional Investor*	*Corporate Banking*
Personalized Offers	*Adaptive financial advice. Real-time savings goal optimization.*	*Loyalty offers can be updated dynamically based don the customer transaction behavior. Insurance policies can be customized.*	*Bespoke investment plans. Dynamic investment portfolios.*	*Optimized loan structures. Custom lending offers. Dynamic pricing.*
Risk and Underwriting	*Real-time risk profiling. Predictive default modeling.*	*Real-time underwriting models.*	*Diversification risk management. Real-time hedging strategies.*	*Real-time risk assessment.*
Operational Efficiency	*Automate routine tasks with context aware workflows.*	*Adaptive Claim management.*	*Automated fund review reports.*	*Streamlined complex operations. Invoice processing, reconciliations.*
Customer Engagement	*Tax and Retirement planning agents. Virtual financial assistants.*	*AI-driven query handling. Proactive policy updates, renewals.*	*Real-time market alerts. Custom research insights.*	*Adaptive tax planning. Financial planning agents.*
KYC/Onboarding	*Adaptive identity verification. Real-time AML compliance.*	*Adaptive identity verification.*	*Real-time due diligence. Investor suitability analysis.*	*Adaptive onboarding workflows. Real-time sanctions monitoring.*
Financial Forecasting	*Future savings, expense insights.*	*Predicting claim reserves. Automated premium calculations.*	*Dynamic investment timing strategies. Investment performance insights.*	*Live cashflow forecasts.*
Fraud Prevention	*Contextual based suspicious activity detection.*	*Claim fraud detection with contextual analysis.*	*Insider trading detection. Automated regulatory compliance.*	*Corporate fraud detection.*

the implementation of Agentic AI, as it will allow users and traders to optimize their investment strategies while managing the volatility of market conditions, (ii) **Healthcare:** Technology advancements can greatly facilitate the improvement of healthcare by supporting users with real-time alerts and by supporting the healthcare providers with real-time details on patients health condition which can prove life saving. Few of the **Limitations** of Agentic AI implementation are: (i) **Financial AI:** pretrained models needs to be retrained frequently to keep the prediction more accurate and optimized, and (ii) **Healthcare AI:** the data and model needs continuous retuning and performance benchmarking to keep the results accuracy as top priority. Below are the **key points** based on the review of Agentic AI: (i) effective implementation of Agentic AI require a robust data pipeline and quality datasets to train [9], (ii) incorporating human oversight and frequent feedback mechanism improves consistency and compliance of AI implementation, (iii) hybrid AI models combined with traditional patterns improves the quality of the results [9], and (iv) addressing these limitations while applying important lessons learned will be essential for advancing Agentic AI across diverse industries.

VI. BENEFITS AND CHALLENGES OF AGENTIC AI IN FINANCIAL INDUSTRY

A. *Benefits of Agentic AI in Financial Services*

Streamline Operations: Agentic AI increases productivity while reducing human error by automating repetitive tasks like data capture, compliance verification, and trans action processing. This frees up staff members to concentrate on strategic, higher-value work rather than mundane administrative duties [10].

Drive Innovation: Agentic AI enables adaptive financial tools, such as real-time asset management systems and personalized robo-advisors, which improve user experience and performance by dynamically modifying strategies in response to market fluctuations and customer preferences.

Improve Customer Interaction: Agentic AI's capabilities in open banking and embedded finance enable the delivery of AI-powered financial assistants that manage finances, optimize decisions, and match strategies with user objectives and risk tolerance, giving consumers previously unheard-of control over their financial well-being [10].

Proactive Compliance Monitoring: Noncompliance can have serious financial and reputational repercussions in a highly regulated sector. Agentic AI proactively detects risks, ensures accurate reporting to conform to changing legal standards, and continuously monitors regulatory compliance [10].

Autonomous Portfolio Rebalancing: Agentic AI, in contrast to conventional optimization techniques, rebalances portfolios on its own based on current market conditions and the goals of each client. This data-driven strategy improves responsiveness and may perform better than traditional investment methods.

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TABLE II
CORE COMPONENTS OF FINANCIAL-GRADE AGENTIC AI [11]

Component	***Definition***	***Role***	***Tools***	***Example***
Perception Layer: The Eyes and Ears of Financial AI	*Gathers structured and unstructured data from various financial sources, employing tools like APIs for market data feeds and databases for transaction records.*	***Unstructured Data:*** *System Logs, Chatbot Transcripts, Earnings or Investor call documentation, SEC information from filings, client communication and client emails.* ***Databases:*** *Client Know your customer information, Transaction records, client portfolios, and Client risk profile and assessment.* ***APIs:*** *Market data feeds (Bloomberg), CRM platforms (Salesforce), payment gateways.*	***Plaid:*** *Securely accesses banking transaction APIs.* ***Apache NiFi:*** *Used to automate data flows between varied data sources.* ***Snowflake:*** *Provides scalable cloud-based data warehousing.* ***Talend:*** *Supports cross-platform data integration.* ***Fivetran:*** *Integrates multiple data sources into cloud environments.* ***Apache Kafka:*** *Processes high-frequency streaming data (example: trading) at scale.*	*A robo-advisor could manage aggregation of a client's account bank balances, investment portfolio, investment and banking preferences, and real-time market data and can provide customized recommendations.*
Reasoning Engine: The Strategic rules engine of Finance Industry	*Converts raw data into actionable insights using advanced AI techniques like machine learning (ML) and reinforcement learning.*	***Large/Small Language Models (LLMs/SLMs):*** *Analyzes earnings or investor day calls details, transcripts and documentation to analyze sentiment shifts affecting stock prices.* ***Reinforcement Learning (RL):*** *Optimize algorithmic trading for max. Sharpe ratio.* ***Machine Learning (ML):*** *used to predict loan defaults using FICO scores and alternative data (e.g., rental history).*	***TensorFlow/PyTorch:*** *Used to develop time-series models for forecasting and checking the volatility in market.* ***QuantLib:*** *Focused on derivatives pricing for complex financial investment models and financial instruments.* ***Hugging Face Transformers:*** *used to optimize BERT models and to detect risks involved 10-K filings.*	*A Tier-1 bank employs Reinforcement Learning to automate bond trading, trade rebalancing, liquidity, and execution speed.*
Action Executor: Financial Decision and their Outcomes	*Implements AI-driven insights into tangible financial actions, utilizing software and hardware actions for trade executions and adjustments.*	***Software Actions:*** *Executes trades via Alpaca API, updates core banking systems (Temenos T24), and sends fraud alerts via Twilio.* ***Hardware Actions:*** *Predictive demand and forecasting models can be used to adjust ATM cash reserves.*	***Smart Contracts (Ethereum):*** *used to automate loan settlements through blockchain.* ***Alpaca Trading API:*** *Enables programmatic stock trading.* ***UiPath:*** *can Automate Know Your Customer document processing for regulatory and compliance.*	*An AI-driven insurer id empowered to automate approval claims that are under $10,000 using RPA, thereby reducing the processing time from weeks or days to minutes.*
Feedback Loop: Continuous Learning in Finance	*Ensures ongoing improvement by continuously adapting to market shifts and regulatory updates through performance monitoring and model retraining.*	***Performance Monitoring:*** *Tracks P&L for trading bots and false positives/negatives in fraud detection.* **Model Retraining:** Updates credit scoring algorithms with fresh economic data. **Regulatory Auditing:** Ensures AI decisions align with MiFID II, GDPR, and SOX compliance standards.		*A mortgage lender is recommended to retrain their AI weekly with new application data to relook and revise based on the rising interest rates.*

B. Challenges of Agentic AI in Finance

Privacy and Cybersecurity Risks: Agentic AI depends on large datasets to define the privacy and security standards while addressing issues on priority. Additionally it is important aspect to maintain balance between data protection and data personalization [14].

Market Volatility: Lowering barriers to automated trading and decision-making could amplify systemic risks. AI-driven strategies operating in sync may lead to herding behavior, increasing market volatility, and abrupt price swings.

Governance and Regulation: Autonomous AI introduces complex governance challenges. Regulatory frameworks must evolve to ensure accountability, ethical AI use, and bias mitigation, especially in high-stakes areas like credit underwriting. Transparency and oversight are critical for maintaining public trust [14].

Explainability and Transparency: Financial institutions and regulators require clear insights into AI decision-making, particularly in high-risk applications. Policies such as the EU AI Act emphasize responsible AI deployment, liability definition, and the necessity of explainability in financial decisions.

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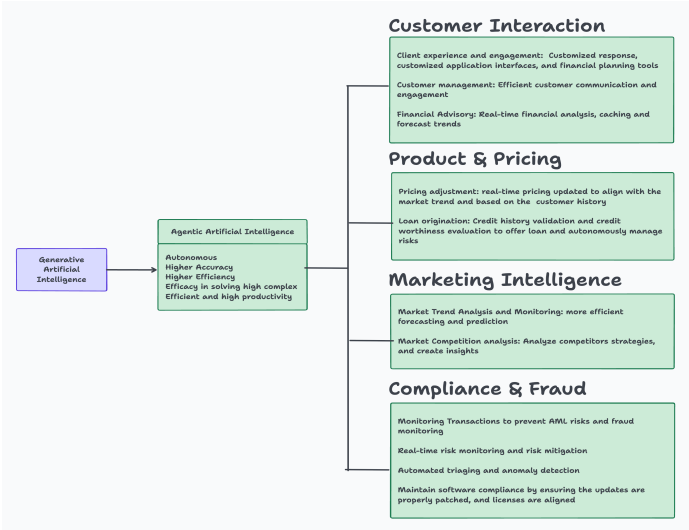

Fig. 3. Agentic Artificial Intelligence Finance Industry Use Cases [19].

VII. The Future of AI in Finance

The evolution of Agentic AI in financial services is a question of when, not if, and how fast the institutions will be able to adapt.. Agents are now introduced as AI mechanisms for working on strategies for trading, portfolio optimisation, and forensic financial analysis, and the sooner the better to get into reality. The distinction will not be between the financial institutes employing AI or not but rather between those that attach it effectively to basic decision-making processes [16].

The key to succeeding in the AI-powered ecosystem is more than just technology adoption. The financial institutions must :(i) Establish AI literacy at all job roles and responsibility levels, with decision-makers conversant with the capacities and limits of Agentic AI,(ii) Strengthen AI governance to promote the alignment of AI initiatives with strategic objectives as well as regulatory compliance, and (iii) Encourage human-machine collaboration where AI is to enhance human financial expertise rather than replacing it.

The firms that have previewed building up AI competency and governance framework will rightly stand to benefit its ability through the transformative potential of Agentic AI.

VIII. Conclusion

Besides improving financial efficiency, Agentic AI holds considerable value for financial inclusion, especially in emerging economies where the traditional banking services go through considerable limitations. FinTechs and banks, with AI-driven automation and intelligent decision making, may cost-effectively scale industries for the undeserved, thus bridging gaps in financial services. To accomplish full use of Agentic AI for many rather than a select few, financial institutions must lay down proper governance frameworks that tie together AI-driven innovation with financial empowerment and inclusion objectives. Ethical AI deployment will be critical to building trust, ensuring fairness, and maximizing AI's positive impact on global finance.

The integration of AI into financial services is entering a new phase from passive assistance to active agency. While traditional AI has operated within predefined constraints retrieving data, summarizing reports, and optimizing workflows the next iteration, Agentic AI, will autonomously plan, execute, and adapt complex tasks with minimal human intervention. This transformation will redefine financial decision-making, accelerating efficiency while introducing new governance challenges that demand proactive oversight.

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